Energy Methods and the Finite Element Method

Practice Guide

AERSP 301, HW 6 topics · a guide to work through with the videos

How to use it. The video pauses at each checkpoint. Work it here, on paper or a tablet, before you press play: the video then works the same step, so you find out at once whether you had it. Stuck for five minutes? Read the hint on page 6. After the last video, two fresh practice problems (page 5) repeat both homework methods with new numbers; their full solutions are at the end.

Notation, as in the videos: U strain energy, W work potential of the loads, $\Pi = U - W$ total potential energy; q_i node displacements; a spring's stretch is (right-end displacement) − (left-end displacement).

Episode 1: Work, Energy and the Bowl

Checkpoint 1.1 The bottom of the bowl

video Ep 1, 6:02

A spring of stiffness $K = 20$ N/mm is fixed to a wall and its free end carries a load $F = 100$ N. Its total potential energy is $\Pi(q) = \frac{1}{2}Kq^2 - Fq$. Write $\Pi(q)$ with the numbers in, then find the q that makes Π smallest, and the value of Π there.

Your answer

Episode 2: Many Springs

Checkpoint 2.1 Two springs in a row

video Ep 2, 6:04

Wall − spring $k_1 = 40$ N/mm − node 1 − spring $k_2 = 20$ N/mm − node 2. Loads $F_1 = 40$ N at node 1 and $F_2 = 20$ N at node 2, both to the right. Solve $\begin{bmatrix} 60 & -20 \\ -20 & 20 \end{bmatrix} \begin{Bmatrix} q_1 \\ q_2 \end{Bmatrix} = \begin{Bmatrix} 40 \\ 20 \end{Bmatrix}$ for q_1 and q_2 (mm).

Your answer

Checkpoint 2.2 Row 3 of the four-node matrix

video Ep 2, 9:51

Node 1 (on the wall) $-$ k_1 $-$ node 2 $-$ two springs k_2 side by side $-$ node 3 $-$ k_3 $-$ node 4. Using the rule "a spring k between nodes i and j adds $+k$ at (i,i) and (j,j) and $-k$ at (i,j) and (j,i)", write row 3 of the 4×4 stiffness matrix.

Your answer

Episode 3: Homework Problem 1

Checkpoint 3.1 The derivative for the rigid bar

video Ep 3, 4:17

For homework problem 1 (a rigid bar, displacement q_2, carries P_2 and P_3; it sits on K_1 and on a K_2–K_2 chain whose junction moves q_1; K_3 joins it to node 3), $\Pi = \frac{1}{2}K_1 q_2^2 + \frac{1}{2}K_2 q_1^2 + \frac{1}{2}K_2(q_2 - q_1)^2 + \frac{1}{2}K_3(q_3 - q_2)^2 - P_1 q_1 - (P_2 + P_3)q_2 - P_4 q_3$. Find $\partial\Pi/\partial q_2$, set it to zero, and collect the terms by q_1, q_2, q_3.

Your answer

Checkpoint 3.2 Test the answer with numbers

video Ep 3, 7:46

The video found $q_2 = \dfrac{P_1 + 2(P_2 + P_3 + P_4)}{2K_1 + K_2}$, $q_1 = \dfrac{P_1 + K_2 q_2}{2K_2}$ and $q_3 = q_2 + \dfrac{P_4}{K_3}$. Put in $K_1 = K_2 = K_3 = 100$ N/mm and $P_1 = P_2 = P_3 = P_4 = 30$ N. Then check one spring with Newton: does K_3 carry exactly P_4?

Your answer

Episode 4: From Springs to Bars

Checkpoint 4.1 **How good is the straight line?** video Ep 4, 6:58

A bar of length L, fixed at $x = 0$ and free at $x = L$, carries a uniform load f per unit length along $+x$. The assumed shape $u = ax$ gave $a = \dfrac{fL}{2EA}$. The exact answer is $u(x) = \dfrac{f}{EA}\left(Lx - \dfrac{x^2}{2}\right)$. Compare the two at the middle, $x = L/2$, in units of fL^2/EA.

Your answer

Episode 5: One Finite Element

Checkpoint 5.1 **An element's stiffness** video Ep 5, 6:19

An aluminum element has $E = 70\,000$ N/mm^2, $A = 100$ mm^2 and length $l = 250$ mm. What is its stiffness EA/l?

Your answer

Episode 6: Assembly and Solving

Checkpoint 6.1 **The column's load vector** video Ep 6, 2:52

A column of height L stands on the ground and carries its own weight, $c = -\rho A g$ per unit length. It is split into three equal elements ($l = L/3$), nodes 1 (base) to 4 (top), with x measured upward (so $c < 0$ means down and $R > 0$ means up). Each element sends $cl/2$ to each of its nodes, and the ground pushes on node 1 with a reaction R. Write the global load vector $\{F_1, F_2, F_3, F_4\}$.

Your answer

Checkpoint 6.2 Stress in element 2

video Ep 6, 7:29

The solution was $\{q_2, q_3, q_4\} = \dfrac{\rho g L^2}{2E}\{-5/9,\ -8/9,\ -1\}$, with $q_1 = 0$. Using $\sigma = E\,(q_{\text{top}} - q_{\text{bottom}})/l$ and $l = L/3$, find the stress in element 2 in units of $\rho g L$.

Your answer

Episode 7: A Stepped Bar and a Spring

Checkpoint 7.1 Assemble four by four

video Ep 7, 1:52

Nodes 0 (left wall), 1 (the step), 2, 3 (right wall). Element 1 joins nodes 0 and 1 with $k_1 = 28\,000$ N/mm; element 2 joins nodes 1 and 2 with $k_2 = 14\,000$ N/mm; a spring $K = 14\,000$ N/mm joins nodes 2 and 3. Assemble the 4×4 global stiffness matrix.

Your answer

Checkpoint 7.2 Stress in the thin piece

video Ep 7, 4:49

Both elements are $l = 500$ mm long; element 1 has $A = 200$ mm^2 and element 2 (the thin piece) $A = 100$ mm^2. With $E = 70\,000$ N/mm^2, $q_1 = 0.2$ mm and $q_2 = 0.1$ mm, find the stress in element 2 (nodes 1 to 2). Tension or compression?

Your answer

Episode 8: Homework Problem 2

Checkpoint 8.1 **Homework 2: K and F** video Ep 8, 2:12

Two elements of $l = 20$ in, each with $EA/l = 1000$ lb/in: node 0 (wall, $x = 0$), node 1 ($x = 20$ in, $P_1 = 50$ lb to the right), node 2 ($x = 40$ in, $P_2 = 100$ lb to the right). A uniform load of 5 lb/in points to the *left* along the whole bar. Assemble the 3×3 global stiffness matrix K and the load vector F (include the wall reaction R at node 0).

Your answer

Checkpoint 8.2 **Homework 2: the reaction** video Ep 8, 4:04

The solution is $q_0 = 0$, $q_1 = 0$, $q_2 = 0.05$ in. Use row 0 of $Kq = F$ (the constrained equation) to find the wall reaction R. Which way does the wall push?

Your answer

After the video: two practice problems

Practice A (springs, like homework problem 1). Wall − spring $k_1 = 20$ N/mm − node 1 − spring $k_2 = 20$ N/mm − node 2 − spring $k_3 = 10$ N/mm − a second wall. A load $F_2 = 60$ N pulls node 2 to the right; node 1 carries no load. (a) Write U, W and Π. (b) Set $\partial\Pi/\partial q_1 = 0$ and $\partial\Pi/\partial q_2 = 0$ and solve. (c) Find the force each wall exerts, and check that they balance the load.

Practice B (finite elements, like homework problem 2). A steel bar, $E = 200\,000$ N/mm^2, $A = 60$ mm^2, $L = 1200$ mm, is fixed at $x = 0$. A uniform load $f = 1.5$ N/mm acts along $+x$, and a tip load $P = 300$ N acts along $+x$. Use three equal elements, nodes 0 (wall) to 3 (tip). (a) Find each element's stiffness and load vector. (b) Assemble K and F. (c) Apply the support and solve for the three free node displacements. (d) Find the stress in each element. (e) Use the constrained equation to find the wall reaction.

Hints

Checkpoint 1.1 Differentiate Π with respect to q and set the slope to zero.

Checkpoint 2.1 Add the two equations: one unknown cancels.

Checkpoint 2.2 Row 3 belongs to node 3. Which springs touch node 3, and what is at the other end of each?

Checkpoint 3.1 q_2 appears in three springs. The chain rule gives a factor -1 where q_2 is subtracted.

Checkpoint 3.2 Work out q_2 first; the other two follow from it.

Checkpoint 4.1 Evaluate both expressions at $x = L/2$.

Checkpoint 5.1 Plug in; watch the units: $\mathrm{N/mm^2} \times \mathrm{mm^2/mm} = \mathrm{N/mm}$.

Checkpoint 6.1 A node shared by two elements receives a share from each.

Checkpoint 6.2 Element 2 runs from node 2 to node 3.

Checkpoint 7.1 Each spring or element drops a 2×2 block on its two nodes; where blocks meet on the diagonal, add.

Checkpoint 7.2 $\sigma = E\,(q_b - q_a)/l$ with q_b the right node.

Checkpoint 8.1 Each element sends $cl/2$, with $c = -5$ lb/in, to each of its nodes; add the point loads at their nodes.

Checkpoint 8.2 Row 0 of K is $1000\,[1,\ -1,\ 0]$, and F_0 includes R; put in $q_0 = q_1 = 0$.

Solutions

Checkpoint 1.1 The bottom of the bowl

$\Pi = 10q^2 - 100q$ (N mm). $d\Pi/dq = 20q - 100 = 0$, so $q = 5$ mm, and $\Pi(5) = 250 - 500 = -250$ N mm. Check: $Kq = 20 \times 5 = 100 = F$, Newton's balance.

Checkpoint 2.1 Two springs in a row

Adding the rows: $40q_1 = 60$, so $q_1 = 1.5$ mm. Row 2: $20(q_2 - 1.5) = 20$, so $q_2 = 2.5$ mm. Check: spring 1 carries $40 + 20 = 60$ N and stretches $60/40 = 1.5$; spring 2 carries 20 N and stretches $20/20 = 1 = q_2 - q_1$.

Checkpoint 2.2 Row 3 of the four-node matrix

Node 3 touches the pair of k_2 springs (to node 2) and k_3 (to node 4). Row $3 = [\,0,\ -2k_2,\ 2k_2{+}k_3,\ -k_3\,]$. It adds up to zero, as every row does before supports are removed.

Checkpoint 3.1 The derivative for the rigid bar

$\dfrac{\partial \Pi}{\partial q_2} = K_1 q_2 + K_2(q_2 - q_1) - K_3(q_3 - q_2) - (P_2 + P_3) = 0$, so $-K_2 q_1 + (K_1 + K_2 + K_3)\, q_2 - K_3 q_3 = P_2 + P_3$.

Checkpoint 3.2 **Test the answer with numbers**

$q_2 = (30{+}2{\cdot}90)/300 = 0.7$ mm; $q_1 = (30{+}70)/200 = 0.5$ mm; $q_3 = 0.7{+}30/100 = 1.0$ mm. K_3 stretches $q_3 - q_2 = 0.3$ mm and carries $100 \times 0.3 = 30$ N $= P_4$. The wall springs carry $K_1 q_2 + K_2 q_1 = 70{+}50 = 120$ N $= 4 \times 30$: everything balances.

Checkpoint 4.1 **How good is the straight line?**

Assumed: $u(L/2) = aL/2 = fL^2/(4EA) = 0.25\, fL^2/EA$. Exact: $\dfrac{f}{EA}\left(\dfrac{L^2}{2} - \dfrac{L^2}{8}\right) = 0.375\, fL^2/EA$. The straight line is too low inside the bar, though it matches at the tip.

Checkpoint 5.1 **An element's stiffness**

$EA/l = 70\,000 \times 100/250 = 28\,000$ N/mm.

Checkpoint 6.1 **The column's load vector**

$F = \dfrac{cl}{2}\{1,\, 2,\, 2,\, 1\} + \{R,\, 0,\, 0,\, 0\} = -\dfrac{\rho A g L}{6}\{1,\, 2,\, 2,\, 1\} + \{R,\, 0,\, 0,\, 0\}$: the shared middle nodes get $cl/2$ from each side.

Checkpoint 6.2 **Stress in element 2**

$q_3 - q_2 = (-8/9{+}5/9)\, \rho g L^2/(2E) = -\frac{1}{3}\, \rho g L^2/(2E)$. Times $E/l = 3E/L$: $\sigma_2 = -\frac{1}{2}\, \rho g L$ (compression). It equals the exact stress $-\rho g L(1 - x/L)$ at the element's middle, $x = L/2$.

Checkpoint 7.1 **Assemble four by four**

$K = \begin{bmatrix} 28\,000 & -28\,000 & 0 & 0 \\ -28\,000 & 42\,000 & -14\,000 & 0 \\ 0 & -14\,000 & 28\,000 & -14\,000 \\ 0 & 0 & -14\,000 & 14\,000 \end{bmatrix}$ N/mm. The overlaps add: $28\,000 + 14\,000$ at node 1 and $14\,000 + 14\,000$ at node 2.

Checkpoint 7.2 **Stress in the thin piece**

$\sigma_2 = 70\,000\,(0.1 - 0.2)/500 = -14$ N/mm$^2 = -14$ MPa: compression.

Checkpoint 8.1 **Homework 2: K and F**

$K = 1000 \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$ lb/in. Each element sends $(-5)(20)/2 = -50$ lb to each node, so $F = \{R - 50,\ -50 - 50 + 50,\ -50 + 100\} = \{R - 50,\ -50,\ 50\}$ lb.

Checkpoint 8.2 **Homework 2: the reaction**

$1000(0) - 1000(0) = R - 50$, so $R = 50$ lb, positive: the wall pushes the bar in $+x$. Check: $50 + 50 + 100 - 5 \times 40 = 0$.

Practice A

(a) $U = \frac{1}{2}(20)q_1^2 + \frac{1}{2}(20)(q_2 - q_1)^2 + \frac{1}{2}(10)q_2^2$ (the right spring's stretch is $0 - q_2$, and squared it is q_2^2); $W = 60\,q_2$; $\Pi = U - W$. (b) $\partial\Pi/\partial q_1 = 20q_1 - 20(q_2 - q_1) = 0$ and $\partial\Pi/\partial q_2 = 20(q_2 - q_1) + 10q_2 - 60 = 0$, i.e. $\begin{bmatrix} 40 & -20 \\ -20 & 30 \end{bmatrix} \begin{Bmatrix} q_1 \\ q_2 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 60 \end{Bmatrix}$. The first row gives $q_2 = 2q_1$; then $-20q_1 + 60q_1 = 60$, so $q_1 = 1.5$ mm, $q_2 = 3.0$ mm. (c) Left wall: $-k_1 q_1 = -30$ N (it pulls the stretched spring back, to the left). Right wall: the right spring is squeezed by 3.0 mm, $10 \times 3 = 30$ N, and the wall pushes left. $60 - 30 - 30 = 0$. Also $\Pi = -90$ N mm $= -\frac{1}{2}(60)(3.0)$.

Practice B

(a) $l = 400$ mm, $k = EA/l = 200\,000 \times 60/400 = 30\,000$ N/mm; each element's load vector is $fl/2 = 1.5 \times 400/2 = 300$ N at each node. (b) $K = 30\,000 \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}$, $F = \{300 + R,\ 600,\ 600,\ 300 + 300\} = \{300 + R,\ 600,\ 600,\ 600\}$ N. (c) Remove row and column 0. Last row: $30\,000(q_3 - q_2) = 600$, so $q_3 - q_2 = 0.02$. Adding it to row 2: $30\,000(q_2 - q_1) = 1200$, so $q_2 - q_1 = 0.04$. Adding both to row 1: $30\,000\,q_1 = 1800$, so $q_1 = 0.06$ mm; then $q_2 = 0.10$ mm, $q_3 = 0.12$ mm. Exact tip: $PL/EA + fL^2/(2EA) = 0.03 + 0.09 = 0.12$ mm. (d) $\sigma = E(q_b - q_a)/l$: $200\,000 \times 0.06/400 = 30$, then 20, then 10 MPa, all tension. (e) Row 0: $30\,000(0 - 0.06) = 300 + R$, so $R = -2100$ N: the wall pulls left with the whole load, $fL + P = 1800 + 300$.